Self-assessment Worksheet (Worksheet 1)

Do I understand this weeks material?

Real Analysis Topics	Tick the box when you feel confident with each topic
• Know the definitions of the number systems $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ as well as their properties.	☐
• Know the definition of the modulus functions and its properties.	☐
• Know the definition of the real numbers $\mathbb{R}$ and its properties.	☐
• Know the definition of the supremum, infimum, maximum and minumum of a set, and know how to prove them using relevant theorems.	☐

Exercises and Questions

Unless stated otherwise, you may assume any results proven in lecture.

Exercises and questions are organised and classified into three groups:

- **Starter Questions** - these questions will help cement the basic techniques.
- **Challenge Questions** - these questions are more challenging, designed to test your deeper understanding of the topic.
- **Extension Questions** - questions to try yourself in your own time to extend your knowledge and understanding.

Starter Qs

Have a go at these "warm-up" questions to get you started.

Question 1 I can do this question ☐

- Find all the values of $x \in \mathbb{R}$ that satisfy the following equalities/inequalities:
  (a) $|x-2| = 10$.
  (b) $|x-4| < 7$.
  (c) $|x+2| < 1$.
  (d) $|x-1||x+1| = 0$.

  If you use any of the properties in Theorem 3.3 of the lecture notes, make sure to explicitly state them.

Question 2 I can do this question ☐

- Prove the reverse triangle inequality from Theorem 3.3 (ix). That is, for every $a, b \in \mathbb{R}$,

$$||a| - |b|| \leq |a - b|.$$

  Hint: You may assume the triangle inequality.

Question 3 I can do this question ☐

- Compute, without proof, the suprema and infima (if they exist) of the following sets:
  (a) $\left\{\frac{m}{n} : m, n \in \mathbb{N} \text{ with } m < n\right\}$.
  (b) $\left\{\frac{(-1)^m}{n} : m, n \in \mathbb{N}\right\}$.
  (c) $\left\{\frac{n}{3n+1} : n \in \mathbb{N}\right\}$.

(d) $\left\{\dfrac{m}{m+n} : m, n \in \mathbb{N}\right\}$.

In each case explain whether or not the supremum is also the maximum of the set or the infimum is also the minumum of the set.

Question 4 I can do this question ☐

- Are the following statements true or false? If they are true, provide a proof. Otherwise, provide a counterexample.
  (a) Every non-empty subset of real numbers has a supremum.
  (b) If M is an upper bound for the set A and $L > M$, then L is also an upper bound for A.

Question 5 I can do this question ☐

- Let A and B be bounded sets of real numbers with $\sup(A) = \sup(B)$ and $\inf(A) = \inf(B)$. Does this imply that $A = B$?

Question 6 I can do this question ☐

- Compute, without proof, the suprema and infima (if they exist) of the following sets:
  (a) $A = \{x \in \mathbb{R} : 3x^2 - 10x + 4 < 1\}$.
  (b) $B = \{x \in \mathbb{Z} : 3x^2 - 10x + 4 < 1\}$.

  In each case explain whether or not the supremum is also the maximum of the set or the infimum is also the minumum of the set.

Question 7 I can do this question ☐

- (a) Prove that $\sqrt{3}$ is irrational.
  (b) What goes wrong when you try to adapt your argument from part (a) to show that $\sqrt{4}$ is irrational?

Question 8 I can do this question ☐

- Prove statement (ii) in Theorem 4.2 of the lecture notes.

Question 9 I can do this question ☐

- Let $A \subset \mathbb{R}$ be a non-empty subset bounded below. Prove that $\inf(A)$ is unique.

Challenge Qs

Now you are feeling more confident try these more challenging questions.

Question 10 I can do this question ☐

- Let $a, b, L \in \mathbb{R}$ such that $|a - L| < r$ and $|b - L| < r$, with $r > 0$. Prove that
  (a) $|2a + 5b - 7L| < 7r$.
  (b) $|3a - 2b - L| < 5r$.

Question 11 I can do this question ☐

- Let $r \in \mathbb{R}$ with $r > 0$ and define $\delta = \min\left\{1, \dfrac{r}{9}\right\}$. Prove that if $|x - 4| < \delta$ then $|x^2 - 16| < r$

Question 12 I can do this question ☐

- Let $\mathbb{I} := \mathbb{R} \setminus \mathbb{Q}$. That is let $\mathbb{I}$ denote the set of irrational numbers.
  (a) Prove that if $a, b \in \mathbb{Q}$, then $a + b \in \mathbb{Q}$ and $ab \in \mathbb{Q}$.
  (b) Prove that if $a \in \mathbb{Q}$ and $r \in \mathbb{I}$, then $a + r \in \mathbb{I}$. Additionally, prove that $ar \in \mathbb{I}$ providing $a \neq 0$.
  (c) The property you prove in part (i) is known as *closure* of multiplication and addition of $\mathbb{Q}$. Do the same properties hold for $\mathbb{I}$? That is, if $r, t \in \mathbb{I}$ that $r + t \in \mathbb{I}$. Similarly, if $r, t \in \mathbb{I}$ is $rt \in \mathbb{I}$. If the statements are true, provide a proof. Otherwise, give counterexamples.

Question 13 I can do this question ☐

- Prove that for every distinct $x, y \in \mathbb{R}$, with $x < y$, there exists a $z \in \mathbb{R} \setminus \mathbb{Q}$ such that $x < z < y$.

  Hint: You may find the density of $\mathbb{Q}$ in $\mathbb{R}$ and the fact that $\sqrt{2}$ is irrational useful.

Question 14 I can do this question ☐

- Prove that every non-empty subset $A \subset \mathbb{R}$ bounded below has an infimum.

Question 15 I can do this question ☐

- In Theorem 3.6, the nested interval property states that the intervals I_n must be closed. That is, if $x \in I_n$ then $a_n \leq x \leq b_n$. Why cannot we use open intervals $I_n = (a_n, b_n)$ instead? Recall that $y \in (a, b)$ means $a < y < b$.

Question 16 I can do this question ☐

- Let $A = \{x \in \mathbb{R} : x < 2\}$. Prove that $\sup(A) = 2$.

Question 17 I can do this question ☐

- Prove that
  (a) $\inf\left(\left\{1 + \frac{1}{n} : n \in \mathbb{N}\right\}\right) = 1.$
  (b) $\sup\left(\left\{\frac{n}{n+1} : n \in \mathbb{N}\right\}\right) = 1.$
  (c) $\inf\left(\left\{\frac{n}{n+1} : n \in \mathbb{N}\right\}\right) = \frac{1}{2}.$

Extension Qs

These questions are optional extension problems so you can gain more practice and confidence with these topics.

Question 18 I can do this question ☐

- Fermat's method of infinite descent is based on the fact that there exists a smallest positive integer, namely 1. If some premise leads to infinitely many integer solutions, where each successive solution is smaller than the last, then this is a contradiction.
  Use Fermat's method of infinite descent to prove that if k is a positive integer and $\sqrt{k}$ is not an integer, then $\sqrt{k}$ is irrational. Follow these steps:
  – Assume that $\sqrt{k}$ is rational, so can be expressed as $\frac{a}{b}$, where a and b are positive integers.

- Let n be the largest integer less than $\sqrt{k}$, and express $\dfrac{a}{b}$ as

$$\frac{a}{b} = \frac{a\left(\sqrt{k} - n\right)}{b\left(\sqrt{k} - n\right)}.$$

- Perform some algebraic manipulations to arrive at a contradiction.

Question 19 I can do this question ☐

- Prove that for every positive number $x \in \mathbb{R}$ and every integer $n \in \mathbb{N}$ there exists a unique positive number $\alpha \in \mathbb{R}$ with $\alpha^n = x$.

End Material

Review and Summary

Make sure you have fully understood and read the notes. Once the solutions for this worksheet have been released, check your answers and make a note of areas of concern. If you are still unsure then email your lecturer or drop by in office hours and we will be happy to help!

Completing Worksheets: Tips

- Proceed carefully and systematically from beginning to end making sure to note theorems and properties utilised.
- Work together with other students, discuss ideas and concepts.
- Prioritise carefulness over speed.
- Include enough information in your answers that you can easily understand your thinking when you come back to review the worksheets at a later date.
- Do ask for help if you get stuck!